



SCI_LAUNCH_A2_W04 · LAUNCH Science

Autumn 2 Week 4 teacher note

Objective (verbatim): Describe DNA, genes & chromosomes.

Scheme of work: LAUNCH scheme of work, Autumn 2 week 4.

Pupil words: I am learning to show how a gene is part of DNA and how DNA is packed into chromosomes.

Reading gate: LAUNCH 11+; rough FK grade 6.2, age proxy ≈ 12.2 , 986 pupil-visible words excluding marked adult guidance. Formula is a rough screen, not an individual reading assessment.

Pearson foundation route: GCSE Biology 1BI0 Paper 1 and Combined Science 1SC0 Biology Paper 1, shared points 3.5 (genome and gene), 3.13 (chromosome and gene terms); shared Foundation. Official specification links: <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-biology-spec.pdf> and <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-combinedscience-spec.pdf>. The workbook's teaching approach is not a new specification point. No Biology-only B/HT point is assumed for Combined Foundation.

Gatsby 4: An NHS genomics laboratory scientist compares gene sections and chromosome changes when interpreting a test.

Sequence, success, retrieval

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40 minutes. Three one-minute resets are inside We do, Independent and Review; Arrival and resets make about eight minutes for settling and transitions.

Success at Starter, checked at Exit: I can label DNA, a gene and a chromosome in a simple model.; I can explain that a gene is a section of DNA with a protein instruction.; I can distinguish one chromosome from the genome.

Retrieval from SoW: "last term" uses the Autumn 1 block in the supplied 2026/27 workbook. If a literal previous academic term is required, staff must check actual teaching rather than invent it.

When	Question	Answer	Exact SoW cell
last lesson	What is one reason a claim of a new stem-cell treatment needs evidence?	To check whether it benefits people and what risks it has.	LAUNCH scheme of work, Autumn 2 week 3
2-4 weeks	Which structure carries the matching chromosome sets after mitosis?	Each new daughter-cell nucleus.	LAUNCH scheme of work, Autumn 2 week 1
last term	Where are chromosomes found in a eukaryotic animal cell?	In the nucleus.	LAUNCH scheme of work, Autumn 1 week 1 (Aut1 W1)

Word meanings and examples (same in lesson and organiser):

Word	Tier	Definition	Example sentence
DNA	3	the molecule that stores genetic instructions	DNA contains the instructions for making proteins.



Word	Tier	Definition	Example sentence
gene	3	a section of DNA that codes for a specific protein	The coloured section of the model stands for one gene.
chromosome	3	a long DNA molecule packaged with proteins in a cell	A chromosome contains many DNA sections, including genes.
genome	3	the entire DNA of an organism	The genome includes all of the organism's chromosomes and other DNA.

Faded example: I do full: Mark a short section of a DNA cord as a gene. Coil the cord as one chromosome; place it inside a border for the genome. Name the section, molecule, package and complete DNA set. We do part-done: DNA cord → a short marked codes for a protein → chromosome package → the genome contains all DNA. Removed step: Label the short marked DNA section “gene” and explain it codes for a specific protein. Independent uses no filled answer. Fade next: Next lesson hide the genome label; pupils reconstruct the whole-set meaning before beginning the Punnett-square topic.

Post-I-do show-me: Show me: is a gene a section of DNA, or is all DNA inside one gene? Expected: A gene is a section of DNA. Every pupil shows a board, point or card. Re-model the named step if insecure.

F/M/H and named stretch

We do F support: Use the full cord model; place the gene label on the short marked section.

We do M: Explain how the short DNA section relates to one gene and the larger package.

We do H stretch: Explain why “one chromosome = one gene” misrepresents scale and role.

Independent F: Label a gene, DNA, chromosome and genome on the model. Stem: This short marked section is a ...; the long package is a ...

Independent M: Explain why a gene is a DNA section, not another material. Stem: A gene is part of ... and codes for ...

Independent H: Correct a faulty gene label and explain the protein instruction. Stem: The faulty label is ...; it should be ... because ...

Go deeper: What distinctions disappear if the whole genome is called one gene?

Named early-finisher stretch — Two-diagram audit: Repair two faulty diagrams: DNA inside a gene, and one chromosome as the entire genome. Justify both. This is a harder correction/evaluation, not a repeated item.

On-screen interactions: Select where each label belongs in the string model and explain one mismatch. uses native selects with a check/reset and aria-live feedback; We do 2 is a native 4-option all-pupil hinge without auto-grading. No storage. Original labelled diagram is simplified, not to scale. Its labels are checked in the fact ledger.

SoW practical status: no core practical is claimed. Prepared model pieces are an optional handling rehearsal with the pupil-facing Risk/Safe lines below.

AfL hinge and misconceptions

Question at We do 2: A new diagram marks a short DNA region carrying instructions for one protein. Which label belongs to that region?



Option	Correct?	Diagnosis
A. A genome containing the whole cell.	no	Treats a short DNA region as the whole genome.
B. A gene within the DNA molecule.	yes	Uses the new case accurately.
C. A complete cell containing no DNA.	no	Labels a DNA segment as an entire cell and overlooks its DNA.
D. A chromosome containing the whole genome.	no	Uses chromosome for the marked gene and confuses one chromosome with the whole genome.

Decision: About 3 in 4 or more correct: proceed; otherwise re-model a coloured section on one long string and ask the parallel hinge. No auto-grade. Parallel hinge: A second diagram outlines an organism's complete DNA set. Which label fits: gene or genome? Answer: genome; it includes all its DNA.

Misconception	How to spot	Correction
A chromosome and a gene are the same whole structure.	Pupil chooses D, calling the marked segment a whole chromosome.	Uncoil the string and point to one marked section for a gene, then the full packaged DNA for a chromosome.
A gene is a protein or a separate cell.	Pupil chooses C, calling the DNA region a complete cell.	The clip is a marker; the short section underneath it is DNA and codes for a protein.

Planned adaptations

Broad area	EEF five-a-day	Concrete action
Cognition & Learning	scaffolding	Scaffolding: Place one completed enlarged string model beside an unlabelled model; fade only the gene marker.
Communication & Interaction	flexible grouping	Flexible grouping: Rehearse "part of" with a partner and allow point/sign/AAC for a labelled explanation.
SEMH	cognitive/metacognitive strategies	Cognitive/metacognitive strategies: Give a private heads-up before critique of a faulty model and a pause choice; return to one label.
Sensory/Physical	technology	Technology: Provide large high-contrast string and an easy-grip clip, or a digital labelled image with speech-to-text.
EAL (language, not SEN)	explicit instruction	Explicit instruction: Use the same four term cards with definition and sentence; allow any-language planning and M/H explanation frames. Keep M and H scientific demand available.

EAL language support remains available at M and H; the challenge stays open. Speak before write; the organiser's "My language" column is blank, without machine translation. Accept point, sign, speak, draw, type, write, exact scribing and staff-captured photo. Offer speech-to-text, TTS and immersive reader.



TA Focus and prompt ladder

At Independent choose at lesson start: teacher works with supported pupils and TA holds group, or TA supports and teacher holds group. Swap across the week. No public cold-calling is required; this is an invitation bank.

I do: "Point to the short marked section, then trace the whole DNA string."

We do: "Where is the missing label on this model? What size is it?"

Independent: "Show the incorrect label first. What does the short section stand for?"

Least prompt first; no physical prompting: Wait five seconds for an independent label. → Point to the marked part of the string. → Ask "Is this marker the entire string or part of it?" → Model the DNA label on one string; pupil labels the coloured part. → Complete one label together, then pupil corrects a fresh diagram alone.

Exit handoff: adult states deepest prompt level, valid response mode and what pupil did independently; other adult chooses the next prompt.

Route	Invitation	Expected answer
F	What is a gene part of?	DNA.
F	What is the name for all an organism's DNA?	The genome.
M	How does a chromosome relate to genes?	It contains a long packaged DNA molecule with gene sections.
M	What does the gene section code for in this specification?	A specific protein.
H	Why is the whole chromosome not correctly labelled as one gene?	A gene is only a section of its DNA, with a coding function.
H	What is lost when genome is used as a synonym for one chromosome?	The genome means the organism's entire DNA, rather than a single package.

Feedback, EHCP broad areas and regulation

Codes exactly: VF · WS · I · NS(+) · E · R · // · ?. VF at We do or Review; WS when supported with recorded prompt level; I when independent at Independent/Exit; NS(+) precise next step at Exit; // a real self-edit at Review; ? adult reads the changed words back on request; R after the pupil responds to NS(+) next lesson and staff judge, never automatic; E only for an actual curriculum observation.

What I record and where: note hinge response split and the named misconception on the staff quick-record grid; record actual code and prompt level in feedback. E: evidence captured on the school's evidence app. Never gather evidence during distress or store pupil data in HTML.

Pathways: high support, VF/WS regular. Build: WS/I mix. Grow: mostly independent, self-edit and response. Launch: independent self-assessment.

Regulation: Check in before the hinge, top rung and Independent. At the agreed pause signal give space and a single return step. Collect no evidence while a pupil is distressed. Re-enter at the last completed step, with one small choice and private correction.

lesson record poster: LAUNCH_SCI_A2_W04. Staff device or projector only; pupil phones not required.

Lundy Exit: pupil can say, point, sign, draw or write; teacher hears the answer and changes the next model/prompt. Four answers: goal at top, actual Review feedback, Exit NS(+), success criteria at Starter and Exit.



Sources, checks and limits

Every scientific and diagram label claim is listed in LESSON_SOURCE.json with a named official source and status. False distractors are deliberately incorrect and diagnosed. Read source claims in context; no medical claim about an individual. The string and SVG are simplified; relative lengths and chromosome shape are not to scale.

Prepared model handling · practical safety

Kit: Short DNA cord, large blunt clips, genome and chromosome frames, tray

Method: Mark one gene with a blunt clip and nest the cord model in its frames.

Pupil Risk: A clip could pinch a finger, and a loose cord could catch.

Pupil Safe: Use large blunt clips and short cord on a tray. Keep the cord away from necks.

Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

Staff handling control — Short DNA cord/string-and-clip DNA model: short length only, counted out and back in, adult-supervised, removed at once if a pupil is dysregulated.

SLT planning and evaluation

Lesson code: SCI-LAUNCH-A2-W04

SoW reference: LAUNCH scheme of work, Autumn 2 week 4

Exact SoW outcome: Describe DNA, genes & chromosomes.

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Evidence in this lesson: Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A2-W04. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Look at this week's string-and-clip DNA model. Point to one detail you notice. Staff: Offer a private, predictable start. Do not require previous knowledge.

Starter (3 min): Recall three ideas from earlier lessons. Show one answer at a time. Staff: All pupils respond; reveal the answers only after attempts. Revisit success criteria.

I Do (5 min): Watch the full model. Point to the step that answers today's question. Staff: Model one complete example; check every pupil's response to the show-me question.

We Do (7 min): Complete the missing step in the part-done model. Speak, then match the cards. Staff: Wait five seconds, gesture, then cue if needed. Support the route without supplying the missing answer.

We Do 2 (5 min): Choose one hinge answer. Show it at the same time as the group. Staff: Collect simultaneous responses, apply the decision rule and parallel hinge in the staff note.

Independent (8 min): Two diagrams label a gene differently. Choose the accurate model and explain one reason. Staff: At lesson start choose one adult with supported pupils and one holding the group; swap during the week. Record independence and prompt level.

Review (4 min): Tell a partner what you changed after feedback. Mark a real edit // and read it back when asked ?. Staff: Give a precise VF or NS(+) only when it actually happens. Invite a true self-edit



and read-back.

Exit (3 min): Show what you know with a model and explain one reason. Staff: Hear the answer privately in a valid mode and use it to change the next example or prompt.

Planned adaptation — cognition learning: Scaffolding: Place one completed enlarged string model beside an unlabelled model; fade only the gene marker.

Planned adaptation — communication interaction: Flexible grouping: Rehearse “part of” with a partner and allow point/sign/AAC for a labelled explanation.

Planned adaptation — semh: Cognitive/metacognitive strategies: Give a private heads-up before critique of a faulty model and a pause choice; return to one label.

Planned adaptation — sensory physical: Technology: Provide large high-contrast string and an easy-grip clip, or a digital labelled image with speech-to-text.

Planned adaptation — eal: Explicit instruction: Use the same four term cards with definition and sentence; allow any-language planning and M/H explanation frames. Keep M and H scientific demand available.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: flexible grouping; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Two diagrams label a gene differently. Choose the accurate model and explain one reason. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W05. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W05: Complete a genetic cross (digital Punnett square).

If behind — dependency: Pass on today’s checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Label the short marked DNA section “gene” and explain it codes for a specific protein. Record the pupil’s own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to “A new diagram marks a short DNA region carrying instructions for one protein. Which label belongs to that region?” in the pupil’s agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Look at this week’s string-and-clip DNA model. Point to one detail you notice. Use the agreed predictable start or pause/re-entry plan; evidence the pupil’s own regulation or engagement target privately.



Sensory/physical: Independent: Two diagrams label a gene differently. Choose the accurate model and explain one reason. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use DNA, gene, chromosome, genome with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.